7. Find the value of k if $f(x) = \begin{cases} kx - 9 & x < 5 \\ 9x - k & x > 5 \\ 36 & x = 5 \end{cases}$ is continuous at $x = 5$.

8. Find the values of m and n, so that given function f is continuous at $x = 3$.

(i) $f(x) = \begin{cases} mx & if\ x < 3 \\ n & if\ x = 3 \\ -2x + 9 & if\ x > 3 \end{cases}$ (ii) $f(x) = \begin{cases} mx & if\ x < 3 \\ x^2 & if\ x \geq 3 \end{cases}$

9. If $f(x) = \begin{cases} \frac{\sqrt{2x+5} - \sqrt{x+7}}{x-2} & , \ x \neq 2 \\ k & , \ x = 2 \end{cases}$

Find the value of k so that f is continuous at $x = 2$.

Review Exercise 2

1. Choose the correct Answer

(i) The function $f(x) = x$ is called ----------

(a) Quadratic function (b) Cubic function
(c) Identity function (d) None of these

(ii) The set $\{x | \ x \in R\}$ can be written as

(a) $[-\infty, \infty]$ (b) $(-\infty, a)$ (c) $(-\infty, \infty)$ (d) (a, ∞)

(iii) $[3,5] - (3,5) =$ ----------

(a) $\emptyset$ (b) $(3,5)$ (c) $[3,5]$ (d) $\{3,5\}$

(iv) Let $R = \{(x, y) | \ x, y \in R \text{ and } y = 5\}$

(a) Set of real numbers (b) $\{5\}$ (c) $\emptyset$ (d) $\{1,5\}$

(v) A function $f(x) = \frac{x}{|x|}, x \neq 0$ is an:

(a) odd function (b) even function
(c) linear function (d) none of these

(vi) If $f(x) = 4^x$ then $f(x+1) - f(x) =$ ----------

(a) 4 (b) $f(x)$ (c) $3f(x)$ (d) $2f(x)$

(vii) The limit of the sequence having general term $a_n = \frac{1}{2^n}$ is:

(a) 0 (b) 1 (c) ∞ (d) None of these

(viii) $\lim\limits_{x \to 0} \frac{\sin ax}{x} = ?$

(a) $\frac{1}{a}$ (b) a (c) ∞ (d) 0

(ix) $\lim\limits_{x \to 3} \frac{x^2 - 9}{x - 3} = ?$

(a) 3 (b) ∞ (c) 0 (d) 6



(x) If $f:[-1,4]\to\mathbb{R}$ is given by $f(x)=x^2$ then $f(-3)$ is:

(a) 9 (b) -9 (c) does not exit (d) 6

(xi) $\lim_{h\to 0}\sin\left(\frac{\pi}{2}+h\right)$

(a) 1 (b) -1 (c) 0 (d) $\frac{1}{2}$

(xii) $\lim_{x\to 0} e^{\frac{-1}{x}}$

(a) 0 (b) 1 (c) ∞ (d) $-\infty$

(xiii) $\lim_{x\to c} f(x)$ exits if and only if

(a) $\lim_{x\to c^+} f(x)$ exist (b) $\lim_{x\to c^-} f(x)$ exist

(c) $\lim_{x\to c^+} f(x)=\lim_{x\to c^-} f(x)$ (d) $\lim_{x\to c^+} f(x)\neq\lim_{x\to c^-} f(x)$

(xiv) $\lim_{n\to\infty}\left[1+\frac{1}{n}\right]^{-7n}$

(a) e^{-7} (b) e (c) 1 (d) ∞

(xv) The limit of the sequence $1, \pi^{-1}, \pi^{-2}, \pi^{-3}, \ldots$ is

(a) 1 (b) π (c) ∞ (d) 0

(xvi) Which of the following represents parametric function

(a) $y=f(x)$ (b) $f(x,y)=0$

(c) $x=f(t), y=g(t)$ (d) None of these

(xvii) If $g(x)=3x+2$ and $g(f(x))=x$ then $f(2)=$ ___

(a) 2 (b) 6 (c) 0 (d) 8

(xviii) The value of k for which the function $f(x)=\begin{cases}\frac{x}{\tan 3x}, & x\neq 0\\ k, & x=0\end{cases}$ is continuous is

(a) 0 (b) 3 (c) $\frac{1}{2}$ (d) $\frac{1}{3}$

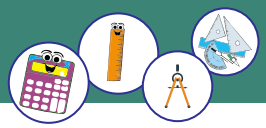
(xix) $\lim_{x\to 0}(1-x)^{\frac{1}{x}}$

(a) e^3 (b) $e^{\frac{-1}{2}}$ (c) e (d) e^{-1}

(xx) $\operatorname{sech}^{-1}x=$ ----------

(a) $\ln\left(x+\sqrt{x^2+1}\right)$ (b) $\ln\left(\frac{1}{x}+\frac{\sqrt{1-x^2}}{x}\right)$

(c) $\frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)$ (d) $\frac{1}{2}\ln\left(\frac{1-x}{1+x}\right)$



$> \frac{d}{dx}(e^x.\sin(x))$

$e^x.\sin(x) + e^x.\cos(x)$

$\frac{1}{(x+1)\sin(x)} - \frac{\ln(x+1)\cos(x)}{\sin^2(x)}$

Exercise 3.6

1. Differentiate the following w.r.t. x:

(i) $\sinh[\ln(x+3)]$ (ii) $\sinh(e^{3x})$ (iii) $\cosh(2x^2+3x)$

(iv) $\frac{\tanh\sqrt{x}}{\sqrt{\cosh x}}$ (v) $\tan(e^{\sinh^{-1}x})$ (vi) $\frac{\sinh^{-1}x}{\text{sech}^{-1}x}$

(vii) $\cosh x\,.\coth x^2$ (viii) $\sinh x\tanh x^2$ (ix) $\ln[\tanh(x^2+2x+1)]$

2. Find $\frac{dy}{dx}$, for the following functions:

(i) $y = x\cosh^{-1}x - \sqrt{x^2-1}$ (ii) $y = x\tanh^{-1}(3x)$

(iii) $\ln(\cosh^{-1}x) + \sinh^{-1}y = C$ (iv) $y = \ln(1-x^2) + 2x\tanh^{-1}x$

(v) $y = \tanh^{-1}(\tan x^3)$ (vi) $y = x\,\text{sech}^{-1}(\sqrt{x})$

3. Write MAPLE command **diff** to differentiate the following:

(i) $f(x) = 2x^3 + 3x^2 + 6$ (ii) $f(x) = \sin(2x+3)$

(iii) $f(x) = (x+1)(x+2)$ (iv) $f(x) = \frac{x^2-3x+2}{x^2-4}$

4. Write MAPLE command $> \frac{d}{dx}$ to differentiate the following functions:

(i) $f(x) = x^3 + 5x^2 + 3x + 7$ (ii) $f(x) = \sin x^2$

(iii) $f(x) = \frac{\sqrt{x}+1}{x^2+1}$

Review Exercise 3

1. Select the correct options:

(i) The derivative of $\frac{2}{x^3}$ is:

(a) $\frac{2}{3x^2}$ (b) $-\frac{2}{3x^2}$ (c) $\frac{6}{x^4}$ (d) $-\frac{6}{x^4}$

(ii) The derivative of $\sqrt{x} + x\sqrt{x}$ is:

(a) $\frac{1}{2\sqrt{x}} + \frac{3\sqrt{x}}{2}$ (b) $-\frac{1+x}{2\sqrt{x}}$ (c) $\frac{1}{\sqrt{x}} + 3\sqrt{x}$ (d) $\frac{1}{\sqrt{x}} + \frac{2\sqrt{x}}{3}$



(iii) If $y = (x+1)(x^2-2)$, then $\frac{dy}{dx}$ is:

(a) $x^3 + x^2 - 2x - 2$ (b) $3x^2 + 2x - 2$ (c) $3x^2 - 2x + 2$ (d) $3x^2 + 2x + 2$

(iv) If $ax^2 + by^2 = ab$, then $\frac{dy}{dx}$ is:

(a) $\frac{-2ax}{by}$ (b) $\frac{-bx}{ay}$ (c) $\frac{-ax}{by}$ (d) $\frac{-ax}{2by}$

(v) If $y = \sqrt{\tan x - y}$, then $\frac{dy}{dx}$ =?

(a) $\frac{\tan x}{2y+1}$ (b) $\frac{-\csc^2 x}{2y-1}$ (c) $\frac{\sec^2 x}{2y-1}$ (d) $\frac{\sec^2 x}{2y+1}$

(vi) If $y = \tan^{-1}\sqrt{x}$ then $\frac{dy}{dx}$ =?

(a) $\frac{1}{1+x^2}$ (b) $\frac{1}{x+\sqrt{x}}$ (c) $\frac{1}{2(x+x\sqrt{x})}$ (d) $\frac{1}{2(\sqrt{x}+x\sqrt{x})}$

(vii) The derivative of $\tan x$ w.r.t. $\cot x$ is:

(a) $\sec^2 x \csc^2 x$ (b) $-\tan^2 x$ (c) $\frac{\sec^2 x}{\csc^2 x}$ (d) $\tan^2 x$

(viii) The $f(x) = ax^2 - 3x - 5$ and $f'(2) = 9$, then a is equal to:

(a) -2 (b) 3 (c) 4 (d) 5

(ix) The derivative of x^2e^{2x} is:

(a) $x^2e^{2x} + 2x^2e^x$ (b) $2xe^{2x}$ (c) $2e^{2x}(x^2+x)$ (d) $2e^{2x}(x^2+1)$

(x) The derivative of a^x, if $a < 0$ is:

(a) $-a^x \ln a$ (b) $a^x . \ln a$ (c) $\frac{a^x}{\ln a}$ (d) Does not exist

(xi) If $y = \tan^{-1}\sqrt{\frac{1-\cos 2x}{1+\cos 2x}}$ then $\frac{dy}{dx}$ =?

(a) 1 (b) -1 (c) 2 (d) $\frac{1}{2}$

(xii) $\frac{d}{dx}(\sinh^{-1} x + \cosh^{-1} x)$ is:

(a) $\cosh^{-1} x - \sinh^{-1} x$ (b) $\frac{1}{\sqrt{1+x^2}} - \frac{1}{\sqrt{1-x^2}}$

(c) $\frac{1}{\sqrt{x^2+1}} + \frac{1}{\sqrt{x^2-1}}$ (d) $\frac{1}{\sqrt{x^2-1}} - \frac{1}{\sqrt{1+x^2}}$

(xiii) The derivative of $\tanh ax$ is:

(a) $\operatorname{sech}^2 ax$ (b) $a \operatorname{sech} ax$ (c) $a \operatorname{sech}^2 ax$ (d) $2a \operatorname{sech}^2 ax$

(xiv) The derivative of $\coth^{-1}(2x)$ is:

(a) $\frac{1}{1-4x^2}$ (b) $\frac{2}{1-4x^2}$ (c) $\frac{2x}{1-4x^2}$ (d) $\frac{2}{1-x^2}$

(xv) If is the function with rule $f(x) = \ln 2x$ $(x > 0)$, if g is the inverse of f, then $g'(x) =$

(a) $\frac{2}{x}$ (b) $\frac{1}{2x}$ (c) $\frac{2}{e^x}$ (d) $\frac{e^x}{2}$

(xvi) If $f(x) = a \cos 3x$ and $f'\left(\frac{\pi}{2}\right) = 6$, then $a =$

(a) -6 (b) -2 (c) 2 (d) 3

2. Find the derivative of $\sqrt{\cos x}$ and $\sec\sqrt{x}$ by first principle.

3. If $y = (\sin x)^{\ln x}$, find $\frac{dy}{dx}$.

4. Find $\frac{dy}{dx}$, if $ax^2 + 2hxy + by^2 = 0$.

5. Let $f(x) = \cot^{-1}\left(\frac{2x}{1-x^2}\right)$, find $f'(x)$ and $f'(-\sqrt{3})$.

6. $x = 4(t - \sin t)$ and $y = 4(1 + \cos t)$, find $\frac{dy}{dx}$.

7. Differentiate w.r.t. x:

(i) $\frac{x^2+x^{-1}}{x^2-x^{-1}}$ (ii) $\frac{3x-2}{\sqrt{x^2+1}}$

8. If $y = x^4 + 2x^2$, show that $\frac{dy}{dx} = 4x\sqrt{y+1}$.

9. If $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \cdots}}}$, show that $(2y - 1)y' = \cos x$.

10. Differentiate w.r.t. x:

(i) $\cosh(\cos^{-1}\sqrt{x})$ (ii) $\tanh^{-1}(\cos 2x)$.

Exercise 4.4

1. Show that function $f(x) = -x^2 + 10x + 9$ is increasing at $x = 4$.
2. Show that $f(x) = \tan^2 x$ is decreasing at $x = \frac{3\pi}{4}$.
3. Find the maximum and minimum values, if any, of the function $f: \mathbb{R} \to \mathbb{R}$ in the following cases:

   (i) $f(x) = x^2 - 2x + 3$ (ii) $f(x) = x^3 - 9x^2 + 15x + 3$

   (iii) $f(x) = -x^4 + 2x^2$ (iv) $f(x) = e^x \sin x$

   (v) $f(x) = 2e^x + e^{-x}$ (vi) $f(x) = 2x - x^2$
4. A rectangular reservoir with a square bottom and open top is to be lined inside with lead. Find the dimensions of the reservoir to hold $\frac{1}{2}a^3$ cubic metres, such that the lead required is minimum.
5. Find a right-angled triangle of maximum area with a hypotenuse of length h.
6. A particle moves so that its distance s at time is given by $s = ut + \frac{1}{2}at^2$, where u a are fixed real numbers. Find its speed and magnitude of its accelerations at time t.

Review Exercise 4

1. Choose the correct answer

(i) If $y = \frac{1}{x}$, then $y'''(1) =$ ----------

(a) 6 (b) -6 (c) 2 (d) None of these

(ii) What will be the nth derivative of $2e^x$

(a) $2ne^x$ (b) $2e^{nx}$ (c) $2e^x$ (d) $\frac{e^x}{2}$

(iii) If $xy = k^2$, then $y'' =$ ___

(a) $\frac{2k^2}{x}$ (b) $\frac{-2k^2}{x^3}$ (c) $\frac{3k^2}{x^3}$ (d) $\frac{2k^2}{x^3}$

(iv) 3rd order derivative of 2^x is ----------

(a) $(ln3)^2\, e^{xln2}$ (b) $(ln2)^3\, e^{xln3}$ (c) $(ln2)^3\, e^{xln2}$ (d) None of these

(v) 2nd order derivative of $f(x) = A\sin x + B\cos x$ is ----------

(a) $f(x)$ (b) $-f(x)$ (c) $\pm f(x)$ (d) None of these



(vi) If S is the distance covered by a car, then $\frac{d^2S}{dt^2}$ will be its ----------

(a) Velocity (b) deceleration (c) acceleration (d)Average velocity

(vii) $y = \tan^{-1}x$, then $y''(1) =$ ----------

(a) $\frac{-1}{2}$ (b) $\frac{1}{4}$ (c) $\frac{-1}{4}$ (d) $\frac{1}{2}$

(viii) If $y = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$ $x \in R$, then y is the Mclaurin series of

(a) $\sin x$ (b)$\cos x$ (c) e^x (d) None of these

(ix) A function $f(x)$ is said to be a decreasing function when $x_1 < x_2$ and

(a) $f(x_1) < f(x_2)$ (b) $f(x_1) > f(x_2)$

(c) $f(x_1) \geq f(x_2)$ (d) $f(x_1) \leq f(x_2)$

(x) If a function $f(x)$ is such that $f'(c) = 0$ then the point $(c, f(c))$ is called

(a) Maximum point (b) Minimum point

(c) Stationary point (d) Critical point

2. Find 2nd order derivative of $f(x) = \ln(1 + x^2)$

3. Find the second derivative of following parametric functions

$x = 3u^2 + 1$ and $y = 3u^2 + 5u$, Find $\frac{d^2y}{dx^2}$

4. Evaluate the third derivatives of the given functions.

(i) $v(x) = x^3 - x^2 + x - 1$

(ii) $f(x) = 7\sin\left(\frac{x}{3}\right) + \cos(1 - 2x)$

(iii) $y = e^{-5x} + 8\ln(2x^4)$

5. Determine the second derivative of the given functions.

(i) $g(x) = \sin(2x^3 - 9x)$ (ii) $z(x) = \ln(7 - x^3)$

(iii) $q(x) = \frac{2}{(6+2x-x^2)^4}$ (iv) $h(t) = cos^2(7t)$

(v) $2x^3 + y^2 = 1 - 4y$ (vi) $6y - xy^2 = 1$

6. Given that $y = \cos x$; find y'; y''; $y'''$and $y^{(4)}$.

7. Find $\frac{d^2y}{dx^2}$; if $x = 4\sin t$, $y = 5\cos t$.

8. Find the rth derivative of $f(x) = x^n$ where $r \leq n$.

9. Find the Taylor series of the function $x^4 + x - 2$ centered at $a = 1$.

5. A particle moves along the curve $x = 2t^2$ and $y = 4t$. Find the component of velocity and acceleration at $t = 2$ in the direction of $2\hat{\imath} + \hat{\jmath}$.

Review Exercise 5

1. Choose the correct answer.

(i) The domain of the vector function $\vec{r}(t) = t^3\hat{\imath} + \frac{1}{t-1}\hat{\jmath} + \ln(t-2)\hat{k}$, $t \in R$

(a) $\{t > 2, , t \in R\}$ (b) $\{t < 2, , t \in R\}$

(c) $\{t > 0, , t \in R\}$ (d) $\{t > 1, , t \in R\}$

(ii) Is the Function $\vec{F}(t) = \cos t\,\hat{\imath} + \sin 2t\ \hat{k}$ at $t = \pi$ ----------

(a) is not continuous (b) is continuous (c) limit does not exist (d) None of these

(iii) What value of t, vector function $\vec{F}(t) = \frac{1}{t-2}\hat{\imath} + \sin t\ \hat{\jmath}$ is continuous at

(a) $t \neq 2,\ t \in R$ (b) $t > 2,\ t \in R$ (c) $t \leq 2,\ t \in R$ (d) $t \in R$

(iv) If $\vec{u} = t^2\hat{\imath} - e^{2t}\hat{\jmath}$ is a vector function and $h(t) = t^2 + 2t - 2$ is scalar function then $\lim_{t\to 0}[h(t)\,\vec{u}(t)]$ = ----------

(a) $-\hat{\jmath}$ (b) $\hat{\jmath}$ (c) $2\hat{\jmath}$ (d) $-2\hat{\jmath}$

(v) If $\vec{u} = t^2\hat{\imath} - e^{2t}\hat{\jmath}$ and $\vec{v} = -2\hat{\imath} - t^2\hat{k}$ are vector functions then $\lim_{t\to 0}[\vec{u}(t) \cdot \vec{v}(t)]$ = ----------

(a) 0 (b) t^4 (c) $t^2 - e^{2t}$ (d) $-2\,t^2$

(vi) If $\vec{f}(t) = t^2\hat{\imath} - e^{2t}\hat{\jmath}$ then $\vec{f}'(t)$ = ----------

(a) $2t\hat{\imath} - 2e^{2t}\hat{\jmath}$ (b) $2t\hat{\imath} - e^{2t}\hat{\jmath}$ (c) $2t\hat{\imath} - 2te^{2t}\hat{\jmath}$ (d) None of these

(vii) The velocity function of a particle, whose motion is given by $\vec{r}(t) = \frac{1}{2\pi}t\,\hat{\imath} + 3\cos(t)\,\hat{\jmath}$ at time $t = \frac{\pi}{2}$ is

(a) $\frac{1}{2\pi}\hat{\imath} + 3\hat{\jmath}$ (b) $\frac{1}{2\pi}\hat{\imath}$ (c) $\frac{1}{2\pi}\hat{\imath} - 3\hat{\jmath}$ (d) None of these

(viii) The magnitude of velocity of a particle at $t = \pi$, whose motion is given by $\vec{r}(t) = 4\cos(t)\,\hat{\imath} + 4\sin(t)\,\hat{\jmath} + \frac{3}{2\pi}t^2\hat{k}$

(a) $\sqrt{5}$ (b) 5 (c) $\sqrt{5\pi}$ (d) 5π

(ix) The magnitude of acceleration of a particle at $t = \frac{\pi}{2}$, whose motion is given by $\vec{r}(t) = 4\cos(t)\,\hat{\imath} + 4\sin(t)\,\hat{\jmath} + \frac{3}{2\pi}t^2\hat{k}$ is:

(a) $\frac{\sqrt{16\pi^2+9}}{\pi}$ (b) $\sqrt{16\pi^2+9}$ (c) $\frac{\sqrt{16\pi^2+9}}{\pi^2}$ (d) $\frac{\sqrt{16+9\pi^2}}{\pi}$



(x) Let $\vec{a} = 2\hat{\imath} + 3\hat{\jmath} + 5\hat{k}$ then $\frac{d\vec{a}}{dt}$ =----------

(a) 10 (b) $\sqrt{38}$ (c) 0 (d) None of these

2. Find the limit of vector function $\vec{r}(t) = 2t\,\hat{\imath} + t^3\,\hat{\jmath} + \hat{k}$ when $t \to 2$.

3. If $\vec{u} = 5\hat{\imath} - 2t\hat{\jmath}$; $\vec{v} = \hat{\imath} - 3t\hat{k}$ are vector functions and $k = t + 1$ is scalar function then find the following:

(i) $\lim\limits_{t\to 0} [\vec{u}(t) - \vec{v}(t)]$ (ii) $\lim\limits_{t\to 1} [\vec{u}(t) \cdot \vec{v}(t)]$

(iii) $\lim\limits_{t\to 1} [\vec{u}(t) \times \vec{v}(t)]$ (iv) $\lim\limits_{t\to 5} [k\,\vec{u}(t)]$

4. Check the continuity of the function $\vec{G}(t) = (t + 8)\,\hat{\imath} + \frac{6}{t-8}\,\hat{\jmath} + \ln(t + 8)\,\hat{k}$ at $t = 0$.

5. For what value of t, following vector functions are continuous

$\vec{r}(t) = \sqrt{36 - t^2}\hat{\imath} + \ln(t + 4)\,\hat{\jmath}$

6. Find $\vec{f}'(t)$ of the following vector functions.

(i) $\vec{f}(t) = e^{t^4}\hat{\imath} + (\mathrm{t}^3 + 3)\hat{\jmath} + \operatorname{cosec} \mathrm{t}^2\,\hat{k}$

(ii) $\vec{f}(t) = \frac{1}{t}\hat{\imath} + e^{2t^3}\hat{\jmath} + \sec \mathrm{t}^3\,\hat{k}$

7. A particle moves along the curve whose parametric equations are $x = t^3 + 2t$, $y = 3e^{-2t}$, $z = 2\sin(5\,t)$, where x, y and z show variations of the distance covered by the particle in cm with time in seconds. Find the magnitude of the acceleration of the particle at $t = 0$.

8. The path of a particle is given for time $t > 0$ by the parametric equations $x = t + \frac{2}{t}$ and $y = 3t^2$. Find velocity of particle when time $t = 1$ and acceleration at $t = 2$.

12. $x^2 + y^2 = 36$ $x = -1\ ,\ x = 1$

13. $\frac{x^2}{4} + \frac{y^2}{9} = 1$ $x = -1\ ,\ x = 1$

14. $y^2 = 2x + 5$ $x = 1\ ,\ x = 2$

15. $y^2 = \tan^4 x$ $x = \frac{\pi}{6}\ ,\ x = \frac{\pi}{4}$

16. Write MAPLE Command to find integration of the following functions:

(i) $f(x) = e^{2x}$ (ii) $f(x) = \sin x$

(iii) $f(x) = \cos 2x$ (iv) (iv) $f(x) = \ln(1 + x)$ (v) $f(x) = \frac{1}{x}$

Review Exercise 6

1. Choose the correct option.

(i) $\int f^n(x) f'(x) dx$, where $n \neq -1$, is

(a) $\frac{f^{n+1}(x)}{n+1}$ (b) $\frac{f^{n-1}(x)}{n-1} + c$

(c) $nf^{n-1}(x) + c$ (d) $\frac{f^{n+1}(x)}{n+1} + c$

(ii) $\int f^n(x) f'(x) dx$, where $n = -1$, is

(a) $\frac{f^{n+1}(x)}{n+1} + c$ (b) $\ln|f^n(x)| + c$

(c) $\ln|f(x)| + c$ (d) $nf^{n-1}(x) + c$

(iii) $\int x^n dx$, where $n = -1$ is

(a) $\frac{x^{n+1}}{n+1} + c$ (b) $nx^{n-1} + c$

(c) $\frac{x^{n-1}}{n-1} - c$ (d) $\ln x + c$

(iv) $\int \sin x \cos x\, dx =$

(a) $\sin x + c$ (b) $\cos x + c$

(c) $\frac{1}{4}\cos 2x + c$ (d) $-\frac{1}{4}\cos 2x + c$

(v) $\int x^2 \ln e^{x^2} dx =$

(a) $\frac{x^2}{4} + c$ (b) $\frac{x^5}{5} + c$

(c) $\ln e^{x^2} + c$ (d) $\ln x^x + c$

(vi) $\int e^{\ln x^3}\, dx =$

(a) $e^{x^3} + c$ (b) $\frac{x^3}{3} + c$

(c) $\frac{x^4}{4} + c$ (d) $\ln x^3 + c$

(vii) $\int (1 + \tan^2 x)\, dx =$

(a) $\tan x + c$ (b) $\sin^2 x + c$

(c) $\frac{\tan^2 x}{2} + c$ (d) $\ln \sec x + c$

(viii) $\int \frac{2e^x}{1+e^x} dx =$

(a) $\ln(1 + e^x) + c$ (b) $\ln(1 + e^x)^2 + c$

(c) $(1 + e^x)^{-2} + c$ (d) $\frac{e^x}{2} + c$

(ix) $\int \frac{e^{x+3\ln x}}{x^3} dx =$

(a) $\frac{1}{3} e^{x+3\ln x} + c$ (b) $e^x + c$

(c) $e^{x+3\ln x} + c$ (d) $3 \ln x + c$

(x) $\int \ln\left(e^x \cdot e^{\sin x}\right) dx =$

(a) $\frac{1}{e^{x+\sin x}} + c$ (b) $\ln \sin x + c$

(c) $\frac{x^2}{2} - \cos x + c$ (d) $x \ln \sin x + c$

(xi) $\int \frac{\frac{1}{x\sqrt{x^2-1}}}{cosec^{-1} x} dx =$

(a) $\ln(\text{cosec}^{-1} x)^{-1} + c$ (b) $(\text{cosec}^{-1} x)^2 + c$

(c) $\text{cosec}^{-1} x + c$ (d) $\ln(\text{cosec}^{-1} x) + c$

(xii) If $F(x)$ is an antiderivative of $f(x)$ then $\int_a^b f(x)\, dx =$

(a) $F(a) - F(b)$ (b) $F(b) - F(a)$

(c) $f(b) - f(a)$ (d) $\frac{f(b)}{f(a)}$

(xiii) $\int_{-50}^{+50} (x^3 + x)\, dx$

(a) 0 (b) 1000 (c) 2000 (d) 3000

(xiv) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^9 x \cos^{11} x\, dx$

(a) 1 (b) 3 (c) 0 (d) $2\int_0^{\frac{\pi}{2}} \sin^9 x \cos^{11} x\, dx$

(xv) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^{10} x \cos^{11} x\, dx$

(a) 0 (b) 1 (c) 3 (d) $2\int_0^{\frac{\pi}{2}} \sin^{10} x \cos^{11} x\, dx$

(xvi) Area bounded by the curve $y = \ln e^{x^2}$ from $x = -1$ to $x = 1$ is

(a) $\frac{2}{3}$ (b) 1 (c) $\ln 2$ (d) $\ln 3$

(xvii) $\int_a^b f(x)\, dx =$

(a) $-\int_a^b f(x)\, dx$ (b) $\int_b^a f(x)\, dx$

(c) $-\int_b^a f(x)\, dx$ (d) 0

(xviii) $\int_2^2 \left(x^3 + 3x^2 - 5x^{-\frac{1}{2}}\right) dx =$

(a) 0 (b) $12 - 10\sqrt{2}$

(c) $24 - 20\sqrt{2}$ (d) $20\sqrt{2}$

(xix) $\int_{-2}^{2} (x^5 - x^3 + x)^5 (5x^4 - 3x^2 + 1) dx =$

(a) $\frac{1}{3}(26)^6$ (b) 0

(c) $2(26)^6$ (d) $\frac{(26)^6}{6}$

(xx) $\int \frac{dx}{a^2+x^2} =$

(a) $tan^{-1}\frac{x}{a}$ (b) $\frac{1}{a}\sec^{-1}\frac{x}{a}+c$

(c) $\frac{1}{a}\tan^{-1}\frac{x}{a}+c$ (d) $\sin^{-1}\frac{x}{a}+c$

2. Evaluate the following integrals.

(i) $\int \frac{5x^2-3x+1}{x^2(x+1)}dx$ (ii) $\int \frac{dx}{\sin x \cos x}$

(iii) $\int \frac{e^{3x}}{1+e^{6x}}dx$ (iv) $\int \frac{(lnx)^7}{x}dx$

(v) $\int \frac{5\cos x dx}{6+\sin x-\sin^2 x}$ (vi) $\int x\sec^2 x\,dx$

(vii) $\int \frac{d\theta}{\cot\theta-\tan\theta}$ (viii) $\int \frac{x}{1+\sqrt{x}}dx$

(ix) $\int \frac{3+3\tan^2 x}{1+9\tan^2 x}dx$ (x) $\int \cot^{-1}2x\,dx$

(xi) $\int \frac{1-\sqrt{x}}{1+\sqrt{x}}dx$ (xii) $\int \sin^5 x\sqrt{\cos x}\,dx$

$$\tan\theta = \frac{2\sqrt{h^2-ab}}{a+b}$$

$$\tan\theta = \frac{2\sqrt{\left(\frac{13}{2}\right)^2 - 5(-6)}}{5-6} = -\frac{17}{6}$$

$$\theta = \tan^{-1}(17) \Rightarrow \boxed{\theta = 86.63°}$$

Exercise 7.8

1. Find the equations of the pair of lines represented jointly by each of the following equations. State nature of lines and also trace the pair of lines;

(i) $x^2 - 5xy + 6y^2 = 0$ (ii) $4x^2 - xy - 5y^2 = 0$

(ii) $9x^2 - 6xy + y^2 = 0$ (iii) $10x^2 - 3xy - y^2 = 0$

(v) $7x^2 - 3xy + 5y^2 = 0$

2. Find the combined equation of the pair of lines through the origin which are perpendicular to the lines represented by

(i) $2x^2 - 5xy + y^2 = 0$ (ii) $6x^2 - 13xy + 6y^2 = 0$

3. Trace the pair of lines given by the following equations;

(i) $x^2 - 3xy + 2y^2 = 0$ (ii) $x^2 - 6xy + 9y^2 = 0$

(iii) $6x^2 - xy - y^2 = 0$ (vi) $8x^2 - 3xy - y^2 = 0$

4. Find the angle between the lines represented by;

(i) $x^2 - 5xy + 6y^2 = 0$ (ii) $3x^2 + 7xy + 2y^2 = 0$

(iii) $x^2 + 2xy - 3y^2 = 0$ (iv) $x^2 + xy - 2y^2 = 0$

5. The gradient of one of the lines of $ax^2 + 2hxy + by^2 = 0$ is twice that of the other. Show that $8h^2 = 9ab$.

Review Exercise 7

1. Select correct option.

(i) The slope of the line that passes through $(-3,-4)$ and $(7,6)$ is

(a) 0 (b) undefined (c) -1 (d) 1

(ii) A line with a slope of $\frac{1}{2}$ and a y-intercept 7 is

(a) $2y = x - 7$ (b) $y = \frac{1}{2}x + 7$

(c) $y = \frac{1}{2}x - 7$ (d) $x - 2y = 14$



(iii) Two lines are said to be parallel if and only if their slopes are

(a) Equal (b) Unequal

(c) Does not exit (d) negative reciprocals of each other

(iv) Two lines l_1 and l_2 are said to be perpendicular if and only if

(a) $m_1 m_2 = -1$ (b) $m_1 m_2 = 1$ (c) $m_1 = -m_2$ (d) $m_1 = \frac{1}{m_2}$

(v) The slope of a line that is perpendicular to a vertical line is

(a) 0 (b) 1 (c) 90° (d) undefined

(vi) The slope of the line which makes an angle 45° with the line $3x - y = -5$ is

(a) 2 (b) $\frac{1}{2}$ (c) $\frac{1}{2}, -2$ (d) -2

(vii) The point on the line $2x - 3y = 5$ is equidistant from $(1, 2)$ and $(3, 4)$ is

(a) $(-2, 2)$ (b) $(4, 1)$ (c) $(1, -1)$ (d) $(4, 6)$

(viii) In a plane three or more points are said to be collinear if

(a) they lie on a circle (b) they form a closed loop together

(c) they lie on a straight line (d) they do not make any defined shape

(ix) If the line coincides with x-axis then its equation is

(a) $y = b$ (b) $-b$ (c) $y = 0$ (d) ∞

(x) The general equation of line also known as standard equation of line is,

(a) $ax + by + c = 0$ (b) $y = ax + c$

(c) $y - y_1 = m(x - x_1)$ (d) $\frac{x}{a} + \frac{y}{b} = 1$

2. If the distance between the points $(5, -2)$ and $(1, a)$ is 5, find the values of a.

3. $M(3, 8)$ is the midpoint of the line AB. A has the coordinates $(-2, 3)$, find the coordinates of B.

4. The diameter of a circle has endpoints: $(2, -3)$ and $(-6, 5)$. Find the coordinates of the center of this circle?

5. Find the coordinates of the points which divides the join of $P(-1, 7)$ and $Q(4, -3)$ in the ratio $2 : 3$.

6. Find the points of trisection of the line segment AB, where $A(-6, 11)$ and $B(10, -3)$.

7. Two vertices of a triangle are $(1, 4)$ and $(3, 1)$. If the centroid of the triangle is the origin, find the third vertex.

8. Find the slope of the line which is perpendicular to the given line whose equation is $-2y = -8x + 9$.

9. If a straight line intercepts the x-axis at $(6,0)$ and intercepts the y-axis at $(0, 5)$, write the equation ofthe straight line in two intercept form.

$\because$ $m_1 m_2 = -1$

$\therefore$ Tangent to the circle is perpendicular to the radial segment at the point of contact.

Exercise 8.4

Prove the following analytically

1. The tangents drawn at the ends of a diameter of a circle are parallel.
2. A normal to a circle passes through the centre of circle.
3. The mid-point of hypotenuse of a right triangle is the centre of the circle circumscribing the triangle.
4. Measure of the central angle of a major arc is double the measure of the inscribed angle of corresponding minor arc.
5. The parallelogram circumscribing a circle is a rhombus.

Review Exercise 8

1. Tick the correct option.

(i) If a plane cuts one nappe of a right circular cone perpendicularly then conic is -----

(a) parabola (b) circle (c) ellipse (d) hyperbola

(ii) The centre of circle with equation $(x+3)^2 + (y-5)^2 = 36$ is ----------

(a) $(3,-5)$ (b) $(-3,-5)$ (c) $(-3,5)$ (d) $(3,5)$

(iii) The centre of circle with equation $x^2 + y^2 + 10x - 8y + 1 = 0$ is ----------

(a) $(-5,\ 8)$ (b) $(-10,\ 8)$ (c) $(5,-4)$ (d) $(-5,\ 4)$

(iv) The radius of circle with equation $x^2 + y^2 + 4x + 6y + 1 = 0$ is ----------

(a) $\sqrt{13}$ (b) $\sqrt{12}$ (c) $\sqrt{10}$ (d) $\sqrt{71}$

(v) Which of the following is a degenerate conic

(a) circle (b) ellipse (c) line (d) parabola

(vi) The equation of circle with centre at origin and diameter of 10 units is ---------

(a) $x^2 + y^2 = 100$ (b) $x^2 + y^2 + 100 = 0$

(c) $x^2 + y^2 = 50$ (d) $x^2 + y^2 = 25$

(vii) For what value of k the radius of circle $x^2 + y^2 + 6x - 4y + k = 0$ is 5

(a) 11 (b) -12 (c) 10 (d) 12

(viii) The centre of the circle $x^2 + y^2 + 6x + 8 = 0$ is:

(a) on x-axis (b) on y-axis (c) in 1st quadrant (d) at origin

(ix) The circle $x^2 + y^2 + 6x + 10y + 9 = 0$

(a) touches x-axis (b) touches y-axis

(c) passes through origin (d) cuts x-axis

(x) The circle $x^2 + y^2 + 20x - 8y + 16 = 0$

(a) touches x-axis (b) touches y-axis

(c) passes through origin (d) cuts y-axis

(xi) The line $y = 2x + c$ will be tangent to $x^2 + y^2 = 25$ if

(a) $c^2 = 25$ (b) $c^2 = 625$ (c) $c^2 = 50$ (d) $c^2 = 125$

(xii) For what value of k, the line $y = 2x + 3$ is tangent to $x^2 + y^2 = k^2$

(a) $\pm\frac{5}{\sqrt{3}}$ (b) $\pm\frac{3}{\sqrt{5}}$ (c) $\pm\sqrt{\frac{3}{5}}$ (d) $\pm\frac{\sqrt{5}}{3}$

(xiii) For what value of k, the line $2x + 3y + k = 0$ is normal to the circle $x^2 + y^2 + 2x + 9 = 0$

(a) -2 (b) 2 (c) 3 (d) -3

(xiv) Equation of tangent to the circle $x^2 + y^2 = 25$ at $(3, 4)$ is:

(a) $3x + 4y = 0$ (b) $4x + 3y = 25$

(c) $3x + 4y = 25$ (d) $3x + 4y = 5$

(xv) Equation of normal to the circle $x^2 + y^2 = 36$ at $\left(2, 4\sqrt{2}\right)$ is:

(a) $2x + 4\sqrt{2}y = 0$ (b) $4\sqrt{2}x + 2y = 0$

(c) $2x - 4\sqrt{2}y = 0$ (d) $4\sqrt{2}x - 2y = 0$

(xvi) The equation of tangent to $x^2 + y^2 = 100$ is _____ if slope of tangent is $\sqrt{15}$

(a) $y = \sqrt{15}x \pm 40$ (b) $y = -\sqrt{15x} \pm 40$

(c) $y = \sqrt{15x} \pm 40y$ (d) $y = -\sqrt{15x} \pm 40y$

(xvii) The length of tangent to the circle $x^2 + y^2 + 2y - 1 = 0$ from $(5, 2)$ is:

(a) $\sqrt{24}$ units (b) $\sqrt{33}$ units (c) $\sqrt{32}$ units (d) $\sqrt{31}$ units

(xviii) Congruent chords of a circle are equidistant from its

(a) diameter (b) centre (c) arc (d) segment

(xix) Angle in a semi-circle is ----------

(a) acute angle (b) obtuse angle (c) right angle (d) straight angle

(xx) The point $(3, 3)$ is _____ the circle $x^2 + y^2 = 64$

(a) outside (b) inside

(c) on (d) cannot be determined

2. Find the equation of circle passing through $(2, 3), (4, 6)$ and

(i) centre on x-axis (ii) centre on y-axis

3. $y = \sqrt{3}x + 10$ is the equation of tangent to the circle with centre at origin. Find the equation of normal to the circle at the point of tangent.

4. Find the condition of tangency, secancy and normality of line $x + y + k = 0$ to the circle $x^2 + y^2 + 2x - 3 = 0$.

3. Find the equation of tangent to the hyperbola $3x^2 - 4y^2 = 12$ when slope is 3.

4. Find the equation of tangent and normal to $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$ at (x_1, y_1).

5. Find the equation of tangent and normal to $\frac{x^2}{5} - \frac{y^2}{7} = 1$ at $\left(2\sqrt{5}, \sqrt{7}\right)$.

6. Find the transformed equation of $\frac{(x-6)^2}{25} + \frac{(y+7)^2}{16} = 1$ when axes are translated with new origin $(6, -7)$.

7. If *xy*-axes are rotated through given angle θ then find the new coordinates of given point P

(i) $(2, 3), \theta = 60°$ (ii) $(6, 7), \theta = 45°$ (iii) $(-4, 6), \theta = 30°$

8. Find new origin O' and new XY-axes with respect to *xy*-coordinate system if it is translated 6 units to the left, 5 units up and rotated $\frac{\pi}{6}$ radians anticlockwise.

9. Find new coordinates of $P(4, 5)$ if new origin is $(1, 2)$ and XY-coordinate system is rotated with $\frac{\pi}{6}$ radians anticlockwise from *xy*-coordinate system.

10. Identity and sketch the curve $xy = 9$.

11. Through which angle the axes be rotated about origin so that the transformed equation of $9x^2 + 12xy + 4y^2 - x - y = 0$ does not contain the term involving XY.

Review Exercise 9

1. Tick the correct option.

(i) If the eccentricity is zero, then the conic is ----------

(a) parabola (b) ellipse (c) circle (d) hyperbola

(ii) The focus of parabola $x^2 = -16y$ is ----------

(a) $(0, 0)$ (b) $(4, 0)$ (c) $(-4, 0)$ (d) $(0, -4)$

(iii) The latus rectum and vertex of $(y - 3)^2 = -8(x + 4)$ is ----------

(a) $-8, (3, -4)$ (b) $8, (3, -4)$ (c) $4, (-3, -4)$ (d) $8, (-4, 3)$

(iv) The equation of tangent at $(4, 6)$ to the parabola $y^2 = 9x$ is ----------

(a) $6y = \frac{9}{2}(x + 4)$ (b) $6y = 9(x - 4)$

(c) $4y = \frac{9}{2}(x + 6)$ (d) $3x - 4y + 12 = 0$

(v) The latus rectum of ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ is ----------

(a) $\frac{5}{32}$ (b) $\frac{32}{5}$ (c) $\frac{50}{4}$ (d) None



(vi) The eccentricity of the conic $\frac{x^2}{5}+\frac{y^2}{4}=1$ is ----------

(a) $\sqrt{5}$ (b) $\frac{1}{\sqrt{5}}$ (c) 5 (d) None

(vii) The centre of ellipse $\frac{(x+5)^2}{10}+\frac{(y-3)^2}{20}=1$ is ----------

(a) $(\sqrt{10},\sqrt{20})$ (b) $(5,3)$ (c) $(-5,3)$ (d) None

(viii) The equation of directrix for the conic $\frac{x^2}{4}+\frac{y^2}{2}=1$ is ----------

(a) $x=\pm\frac{4}{\sqrt{2}}$ (b) $x=\pm\frac{\sqrt{5}}{4}$ (c) $x=\pm\frac{2}{\sqrt{5}}$ (d) $x=\pm 2\sqrt{2}$

(ix) $ax^2+by^2+gx+fy+c=0$ where a, b, g, f and c are real numbers that represents hyperbola if

(a) a and b are non-zero and of same sign

(b) a and b are non-zero and of different sign

(c) either $a=0$ or $b=0$ (d) $a=b=0$

(x) Auxiliary circle of ellipse $\frac{x^2}{6}+\frac{y^2}{5}=1$ is ----------

(a) $x^2+y^2=36$ (b) $x^2+y^2=25$

(c) $x^2+y^2=5$ (d) $x^2+y^2=6$

(xi) The equations of directrices for $\frac{(x-h)^2}{p^2}+\frac{(y-k)^2}{q^2}=1$ are ---------- where $q>p$

(a) $x=\pm\frac{p}{e}$ (b) $x-h=\pm\frac{q}{e}$

(c) $y-k=\pm\frac{q}{e}$ (d) $x-h=\pm\frac{p}{e}$

(xii) The vertices of hyperbola $\frac{x^2}{9}+\frac{y^2}{16}=1$ are ----------

(a) $(\pm 5,0)$ (b) $(0,\pm 5)$ (c) $(0,\pm 4)$ (d) $(\pm 4,0)$

(xiii) Conjugate hyperbola to $\frac{x^2}{5}-\frac{y^2}{6}=1$ is ----------

(a) $\frac{x^2}{5}-\frac{y^2}{6}=1$ (b) $\frac{y^2}{6}-\frac{x^2}{5}=1$ (c) $\frac{x^2}{6}-\frac{y^2}{5}=1$ (d) None

(xiv) The eccentricity of rectangular hyperbola is ----------

(a) 1 (b) 2 (c) $\sqrt{3}$ (d) $\sqrt{2}$

(xv) The equation of tangent to $\frac{x^2}{6}+\frac{y^2}{5}=1$ at $(\sqrt{6},0)$ is ----------

(a) $x=6$ (b) $x=\sqrt{6}$ (c) $y=\sqrt{6}$ (d) None

(xvi) For what value of k, $y=k$ is tangent to the ellipse $\frac{x^2}{25}+\frac{y^2}{9}=1$ is ----------

(a) ± 3 (b) ± 5 (c) $\pm\frac{7}{5}$ (d) None

(xvii) The equation of tangent to $\frac{x^2}{16}-\frac{y^2}{9}=1$ with slope 2 is ----------

(a) $y=2x\pm\sqrt{23}$ (b) $y=2x\pm\sqrt{41}$

(c) $x=2y\pm\sqrt{23}$ (d) $y=2x\pm\sqrt{55}$

(xviii) The equation of $xy=c^2$ represents

(a) parabola (b) ellipse (c) hyperbola (d) circle

(xix) If origin is shifted to $(2,3)$ then coordinates of $(5,6)$ are ----------

(a) $(2,2)$ (b) $(3,3)$ (c) $(4,4)$ (d) None

(xx) If xy-coordinate system is rotated at angle of $\frac{\pi}{4}$ transformation for abscissa is

(a) $x=\frac{x'}{2}-\frac{y'}{\sqrt{2}}$ (b) $x=\frac{x'}{\sqrt{2}}-\frac{y'}{\sqrt{2}}$

(c) $x=\frac{x'}{\sqrt{3}}-\frac{y'}{\sqrt{3}}$ (d) None

2. Find the foci, vertices and directrices for the conic

(a) $\frac{(x-5)^2}{25}+\frac{(y+3)^2}{16}=1$ (b) $\frac{(x+4)^2}{9}-\frac{(y+7)^2}{16}=1$

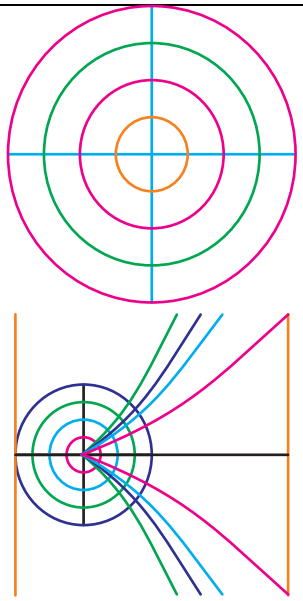
3. Find the condition of tangency the line $y=x+c$ is tangent to the conic

(i) $y^2=10x$ (ii) $2x^2+3y^2=6$

(iii) $5x^2-7y^2=35$

4. Find transformed equation of $\frac{(x+5)^2}{7}-\frac{(y-3)^2}{5}=1$ when new origin is $(-5,3)$.

5. If xy-axes are rotated through angle θ, find coordinates of P if new coordinates is $(-2,7), \theta=45°$.

Maple Command Format	Description
	following solution $1 = 6x^2 + 9y^2, 3 = 6x^2 + 9y^2, 5 = 6x^2 + 9y^2, 9 = 6x^2 + 9y^2$ 12. We also want to graph the original family. i.e., $y^2 = kx^3$ By taking different values of k i.e., 1,3,5,9 we have the following equations $y^2 = x^3, y^2 = 3x^3, y^2 = 5x^3, y^2 = 9.x^3$ 13. we can draw the graphs of these function and drag them on its differential functions graph (ellipses) by using Maple which will add them in. And we'll drag and drop one at a time so we can see that they are curves, these families of curves intersect everywhere at 90 degree angles. And so, the trajectories, the paths of those functions are orthogonal.
$1 = 6x^2 + 9y^2, 3 = 6x^2 + 9y^2, 5 = 6x^2 + 9y^2, 9 = 6x^2 + 9y^2$ $1 = 6x^2 + 9y^2, 3 = 6x^2 + 9y^2, 5 = 6x^2 + 9y^2, 9 = 6x^2 + 9y^2$ $y^2 = 1x^3, y^2 = 3x^3, y^2 = 5x^3, y^2 = 9.x^3$ $y^2 = x^3, y^2 = 3x^3, y^2 = 5x^3, y^2 = 9.x^3$	

Exercise 10.3

1. Find the orthogonal trajectory of the curves $y = ax^2$.
2. Find the orthogonal trajectories of the hyperbola $xy = c$.
3. Find the orthogonal trajectories of the family of parabolas $y^2 = 4ax$.
4. Find the orthogonal trajectories of the family of curves $y = \frac{x}{1+c_1 x}$.
5. Find the general equation of family of curves perpendicular to the $y = c_1 \sin x$.
6. Find the general equation of family of curves perpendicular to the $x^{\frac{1}{3}} + y^{\frac{1}{3}} = c$.

Review Exercise 10

1. Tick the correct answer.

(i) The order and degree of the differential equation $1 + \frac{d^2y}{dx^2} = x\frac{dy}{dx}$ is _____.

(a) order 2, degree 2 (b) order 2, degree 1

(c) order 1, degree 2 (d) order 1, degree 1



(ii) The degree of differential equation $\left(\frac{dy}{dx}\right)^4 + 3\frac{d^2y}{dx^2}$ is _____.

(a) 2 (b) 4 (c) 1 (d) 3

(iii) The order and degree of differential equation $\left(\frac{d^3y}{dx^3}\right)^2 = \sqrt{\frac{dy}{dx}}$ is _____.

(a) order 3, degree 4 (b) order 4, degree 3

(c) order 2, degree 1 (d) order 1, degree 2

(iv) The degree of differential equation $y = x\left(\frac{dy}{dx}\right)^2 + \left(\frac{dy}{dx}\right)$ is _____.

(a) 1 (b) 2 (c) 3 (d) 4

(v) The order and degree of differential equation $\frac{d^2y}{dx^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^3}$ is _____.

(a) 2, 2 (b) 2, 3 (c) 3, 2 (d) 2, 1

(vi) Differential equation $xdy - ydx = 0, y(1) = 2$ has a solution given by y. Then $y(-1)$ is _____.

(a) -1 (b) -2 (c) 2 (d) 1

(vii) The solution of differential equation $\frac{dy}{dx} + y^2 = 0$ is _____.

(a) $y = ce^n$ (b) $y = \frac{1}{x+c}$

(c) $y = -\frac{x^3}{3} + c$ (d) $y = \frac{x^2}{3} + c$

(viii) The general solution of the differential equation $9y\frac{dy}{dx} + 4x = 0$ is _____.

(a) $4x^2 + 9y^2 = c$ (b) $\frac{x^2}{9} - \frac{y^2}{4} = c$

(c) $4x^2 + y^2 = c$ (d) $9x^2 - 4y^2 = 0$

2. Show that $y = Ae^{3x} + Be^{4x}$ is the general solution of the differential equation $\frac{d^2y}{dx^2} - 7\frac{dy}{dx} + 12y = 0$.

3. Solve the following differential equation

(i) $\cos(x + y)\, dy = dx$ (ii) $x^2\frac{dy}{dx} = x^2 + xy + y^2$

(iii) $(xy + y^2)dx = (x^2 - xy)dy$ (iv) $x\frac{dy}{dx} = y + x\tan\frac{y}{x}$

(v) $(2x + 3y - 5)dy + (3x + 2y + 1)dx = 0$

(iii) $f(x,y) = x^3 + 3x^2y + 2y^2x + y^3$ (iv) $f(x,y) = \cos^{-1}\left(\frac{x^2-y^2}{xy}\right)$

(v) $f(x,y) = \frac{x^2-xy+y^2}{x+y^2}$ (vi) $f(x,y) = \sqrt{x^8 - 3x^2y^6}$

(vii) $f(x,y) = x^3 \sin\left(\frac{y^2}{x}\right)$ (viii) $f(x,y) = \ln\left(\frac{x^2+y^2}{x+y}\right)$

4. Verify Euler's theorem for the following homogeneous function.

(i) $f(x,y) = xy + y^2$ (ii) $f(x,y) = \cos\left(\frac{x}{y}\right)$

(iii) $f(x,y) = \sqrt{xy} - x$ (iv) $f(x,y) = \ln\left(\frac{x+y}{y}\right)$

5. If $u = x^2(y-x) + y^2(x-y)$ then show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = -2(x-y)^2$.

6. If $u = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$ then prove that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = \sin 2u$.

7. Use MAPLE command >**diff** or $\left(\frac{\partial}{\partial x} f\right)$ to partial differentiate with respect to x and y of the following functions:

(i) $f(x,y) = x^2y + xy + xy^2$ (ii) $f(x,y) = y + x\cos(y)$

(iii) $f(x,y) = \frac{x+\sqrt{x}}{y+\sqrt{y}}$

Review Exercise 11

1. Multiple choice questions (MCQs)

(i) Given that $f(x,y) = e^{xy}$ then $\frac{f_x}{f_y} =$ _____.

(a) $\frac{x}{y}$ (b) 1 (c) $\frac{y}{x}$ (d) $\frac{y+x\,f'(x,y)}{x+y\,f'(x,y)}$

(ii) Surface area of a cube is a function of _____ variables.

(a) 1 (b) 2 (c) 3 (d) 4

(iii) Given that $g(x,y) = \cos\left(\frac{x}{y}\right)$ then $\frac{g_y}{g_x} =$ _____.

(a) $-\frac{x}{y}$ (b) $\frac{x}{y}$ (c) $-\frac{x}{y^3}$ (d) $-\frac{y}{x}$

(iv) A function $\tan\left(\frac{2x}{3y}\right)$ is a homogeneous function of degree _____.

(a) undefined (b) $\frac{2}{3}$ (c) 1 (d) 0



(v) The perimeter of rectangle is given by a function $P(x,y) = 2(x+y)$, where x and y are respectively its length and breadth. Then sum of partial derivatives w.r.t their independent variables is _____.

(a) $2x$ (b) $2y$ (c) $2(x+y)$ (d) 4

(vi) Given that $z = f(x,y)$ is a homogeneous function of degree 0 then $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} =$ __.

(a) $(x+y)\,f'(x,y)$ (b) $x+y$ (c) 0 (d) $f(x,y)$

(vii) The area of trapezium is a function of _____ variables.

(a) 1 (b) 2 (c) 3 (d) 4

(viii) Given $w = f(u,v)$ is a homogeneous function of degree $\frac{2}{3}$ then $u\frac{\partial w}{\partial u} + v\frac{\partial w}{\partial v} =$ ____.

(a) $(u+v)f'(u,v)$ (b) 0 (c) $\frac{2}{3}$ (d) $\frac{2}{3}w$

(ix) Given that $z = y\left(\frac{u}{v}\right)$ is a homogeneous function of degree 0 then $v\frac{\partial z}{\partial v} =$ _____.

(a) $u\frac{\partial z}{\partial u}$ (b) 0 (c) $-u\frac{\partial z}{\partial u}$ (d) -1

(x) Let $f(x,y)$ and $g(x,y)$ are homogeneous functions of degrees 2 and 3 respectively, then degree of homogeneous functions $\frac{f(x,y)}{g(x,y)}$ is _____.

(a) 6 (b) 1 (c) $\frac{2}{3}$ (d) -1

2. Let $f(x,y) = xy$ and $g(x,y) = \frac{1}{2}xy$ be the homogeneous functions for the areas of rectangle and triangle respectively, where x and y are their independent variables. Are $f+g, f-g, fg$ and $\frac{f}{g}$ homogenous? If yes, what are their degrees?

3. Verify Euler's theorem for the function $z = \sqrt{x^2+y^2}$

4. Given that $z = g(x,y)$ is a homogeneous function of degree 3 then show that $x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = 3z$.

5. Given that $y = f(u,v)$ is a homogeneous function of degree $-\frac{3}{2}$ then show that $u\frac{\partial y}{\partial u} + v\frac{\partial y}{\partial v} = -\frac{3}{2}y$.

6. Given that $u = \sin^{-1}\left(\frac{\sqrt{x}-\sqrt{y}}{\sqrt{x}+\sqrt{y}}\right)$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 0$.

7. Given that $u = \sec^{-1}\left(\frac{x^3-y^3}{x+y}\right)$, show that $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 2\cot u$.



Exercise 12.4

1. Evaluate the following integrals by Trapezoidal rule

(i) $\int_0^2 e^x\,dx$ with 6 intervals (ii) $\int_1^3 \frac{1}{\sqrt{x}}dx$ with 5 intervals

(iii) $\int_0^{\frac{\pi}{2}} \sin x\,dx$ with 7 intervals (iv) $\int_0^{1.4} e^{-x^2}dx$ with 5 intervals

2. Evaluate the following integrals by Simpson $\frac{1}{3}$ rule

(i) $\int_2^3 (4x^2+6)dx$, with 4 intervals (ii) $\int_0^2 \frac{1}{e^x}dx$, with 6 intervals

(iii) $\int_0^{\frac{\pi}{3}} \sqrt{\sin x}\,dx$ with 6 intervals (iv) $\int_0^1 \frac{2}{1+x^2}dx$ with 8 intervals

3. Evaluate the following integrals by Simpson $\frac{3}{8}$ rule

(i) $\int_1^3 \sqrt{x}\,dx$, with 6 intervals (ii) $\int_1^6 \frac{\ln x}{x}dx$, with three sub intervals

(iii) $\int_0^2 \frac{e^{2x}}{1+x^2}dx$ with 9 sub intervals (iv) $\int_0^{\frac{\pi}{4}} \sin x\,dx$ with 6 sub intervals

4. Write MAPLE Command Trapezoid for trapezoidal rule and SIMPSON for Simpson rule

(i) $\frac{x^2-2}{2}, x = 0..1\ \ n = 10$, method trapezoiadal rule

(ii) $\sqrt{9+x^2},\ x = 0..4\ \ n = 10$, method trapezoiadal rule

(iii) $\frac{1}{x^2+4x+3}, , x = 0..2\ \ n = 10$, method simpson's rule

(iv) e^{-x^2} , $x = 0..2\ \ n = 10$, method simpson's rule

Review Exercise 12

1. Select the correct option.

(i) If real root of an equation $f(x) = 0$ lies in the interval $[a, b]$ then $f(a)f(b)$ will be

(a) > 0 (b) < 0 (c) $= 0$ (d) All of them

(ii) In bisection method, the approximate root is a/an _____ of end point of an interval in which actual root lies

(a) Arithmetic mean (b) Geometric mean (c) Sum (d) Product

(iii) Iterative formula for False Position Method to solve the equation $f(x) = 0$ at interval $[a, b]$ is

(a) $\frac{af(a)-bf(b)}{f(a)-f(b)}$ (b) $\frac{af(b)-bf(a)}{a-b}$

(c) $\frac{af(a)-bf(b)}{f(b)-f(a)}$ (d) $\frac{af(b)-bf(a)}{f(b)-f(a)}$

(iv) The fastest method to solve the nonlinear equation numerically is

(a) Bisection Method (b) False Position Method

(c) Newton Raphson Method (d) Both a and b

(v) Newton Raphson Method fails when derivative value of $f(x)$ becomes

(a) > 0 (b) < 0 (c) $= 0$ (d) All of them

(vi) Iterative formula of Newton Raphson method of solve $f(x) = 0$ is _____

(a) $x_{n+1} = x_n + \frac{f'(x_n)}{f(x_n)}$ (b) $x_{n+1} = x_n - \frac{f'(x_n)}{f(x_n)}$

(c) $x_{n+1} = x_n + \frac{f(x_n)}{f'(x_n)}$ (d) $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

(vii) Numerical integration comprises a broad family of algorithms for calculating the numerical value of a?

(a) Definite integral (b) Indefinite integral

(c) Simple integral (d) Compound integral

(viii) In Trapezoidal rule the number of sub interval is the multiple of:

(a) 0 (b) 1 (c) 2 (d) 3

(ix) In Simpson One Third Method, the number of subinterval is the multiple of

(a) 4 (b) 1 (c) 2 (d) 3

(x) The fastest method to solve the definite integral numerically is

(a) Trapezoidal Rule (b) Simpson One Third Rule

(c) Simpson Three Eight Rule (d) Both a and b

2. Using Bisection method find the root of $cosx - xe^x$ with a = 0 and b = 1, by taking 5 iterations.

3. Find a root for the equation $2e^x sinx = 1$ using the false position method and correct it to three decimal places with three iterations, taking $[1, 2]$ as an interval.

4. Find the cube root of 12 using the Newton Raphson method assuming $x_0 = 2.5$.

5. Solve $\int_0^1 cosx^2 dx$ using trapezoidal rule for $n = 5$.

6. Solve $\int_{-2}^4 e^{x^2} dx$ using Simpson one Third as well as Simpson Three Eight rule for $n = 6$.